MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M25, METAL PARTS ASSEMBLY, REMANUFACTURED FROM M20 and M20A1 CARTRIDGE ACTUATED THRUSTERS

1. SCOPE

1.1 This specification covers remanufacture of M20 and M20A1 Cartridge Actuated Thrusters to Thruster, Cartridge Actuated, M25, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION
Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167 - Ammunition Data Card

DRAWINGS
Munitions Command
F10534178 - Thruster, Cartridge Actuated, M25, Metal Parts Assembly
C8797741 - Container, Ammunition, Fiber M347 for Thruster, Cartridge Actuated, M25 or M5A2
C8797742 - Box, Packing, Ammunition for Thruster, Cartridge Actuated M25 or M5A2 in Fiber Container M347

FSC 1377

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IEL-10534178 - Inspection Equipment List for Thruster, Cartridge Actuated M25, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Hydrostatic pressure.- The body and head shall withstand a constant internal pressure of 8000 pounds per square inch (psi) minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

3.3 Structural strength.- The assembled thruster shall withstand a static tensile force of 2000 pounds, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure (see 6.1).

3.4 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled with sufficient torque to bottom components.

3.4.1 All body assembly remanufactured components shall be modified in accordance with instructions of 6.3.

3.4.2 The remanufactured thruster metal parts assembly shall contain a new Piston 10534181, Unlocking Piston 10534182, Piston Locking Spring 10534184, Spring Pin MS171524, Shear Pin 10521618, Ring 10534183, Keys (3) 10534180 and "O" Rings MS29513.

3.4.3 Trunnion rotation.- Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

3.5 Protective finish .- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour condifuration (see 6.3.1).

3.6 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of 6 inches (see 6.2).

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3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- All components shall be free from cracks, splits, or other metal defects.

3.7.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference is made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- The shipping lot of metal parts assemblies shall consist of 600 thrusters, maximum, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.2.2.1 Reclaimed parts.- Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in tables I and II. For an individual defect (single characteristic) the criteria in table I shall be used. For total defects within a major or minor group, the criteria in table II shall be used. All nonconforming material shall be rejected.

Table I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

Table II

	Total Allowable Defects per Sample Size			
Sample Sizes for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Sizes for Minor Defects Number of Defects in Class	**110**	**150**	**225**	**300**
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

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4.2.2.3 Pin, Firing, Drawing D8593864 covering a detail of Drawing F10534178.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp corner	Visual
104. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.4 Body and Pellet Assembly, Drawing D8594481 covering a detail of Drawing F10534178.

CRITICAL	
None defined.	
MAJOR	
101. Concentricity of groove with bore	Gage
102. Concentricity of small counterbore with bore	Gage
103. Concentricity of large counterbore with bore	Gage
104. Diameter of groove	Gage
105. Diameter of small counterbore	Gage
106. Location of groove	Gage
107. Width of groove	Gage
108. Profile of groove	Gage
109. Thread damaged or not full	Visual
110. Metal defective	Visual
111. Protective finish inadequate or defective	Visual
112. Burr or sharp corner	Visual
113. Foreign matter	Visual
114. Pellets loose, missing or improperly inserted	Visual
MINOR	
201. Symmetry of slots with each other and longitudinal axis of body	Gage
202. Diameter of large counterbore	Gage
203. Length from end to shoulder of large counterbore	Gage
204. Length from end to shoulder of small counterbore	Gage
205. Depth of slots	Gage
206. Width of slots	Gage

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Defect	Method of Inspection
207. Radius or chamfer incorrect	Visual
208. Radius or chamfer missing	Visual
209. Surface finish improper except as otherwise specified	Visual
210. Marking missing, incorrect or illegible	Visual

4.2.2.5 Trunnion, Drawing D8594482 covering a detail of Drawing F10534178.

CRITICAL
None defined.

MAJOR

101. Diameter of lugs	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual

MINOR

201. Burr or sharp corner	Visual
202. Foreign matter	Visual

4.2.2.6 Head, Drawing D10521455 covering a detail of Drawing F10534178.

CRITICAL
None defined.

MAJOR

101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual

MINOR
None defined.

4.2.2.7 Retainer, Guide, Drawing D10521616 covering a detail of Drawing F10534178.

CRITICAL
None defined.

MAJOR

101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual

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Defect	Method of Inspection
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
106. Pellets loose, missing or improperly inserted	Visual

MINOR

None defined.

4.2.2.8 Sleeve, Drawing C10521617 covering a detail of Drawing F10534178.

CRITICAL

None defined.

MAJOR

101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual

MINOR

None defined.

4.2.2.9 Assembly, Drawing F10534178, less all component parts except Guide Retainer 10521616, Firing Pin 8593864, O-Ring MS29513-12.

CRITICAL

1. Protrusion of the firing pin	Gage

MAJOR

101. Any part missing	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual

MINOR

None defined.

4.2.2.10 Metal Parts Assembly, Drawing F10534178.

CRITICAL

None defined.

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Defect	Method of Inspection
MAJOR	
101. Trunnion not free to rotate 360 degrees	Manual
102. Any part missing, except as otherwise classified	Visual
103. Improper assembly	Visual
104. Thread damaged or not full	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Shipping cap or plug missing	Visual
202. Protective finish inadequate or defective	Visual

4.2.2.11 Container, Fiber, Drawing C8797741.

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Gap between body and cover, max	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.12 Box, Packing, Drawing C8797742.

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual

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4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Hydrostatic pressure.- Each body and each head shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.2 shall be cause for rejection of the unit.

4.2.3.2 Structural strength.- Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.3 shall be cause for rejection of the thruster.

4.2.3.3 Protective finish.- The number of specimens and acceptance criteria protective finish tests shall be in accordance with the applicable specifications listed on the drawings.

4.2.3.4 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.6 shall be cause for rejection of the assembly.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10534178.

4.3 Test methods and procedures.

4.3.1 Hydrostatic pressure.- Prior to application of protective finish, the body and head, individually, shall be mounted in an approved device. The testing device used shall internally seal the body and head, as applicable, at each end without applying any axial load other than that applied by the liquid pressure.

4.3.2 Structural strength.- The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.3. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.3.3 Protective finish.- Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.3.4 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster six inches from and at the same height as an approved compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading as shown in figure 1. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8797741 and C8797742.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 When remanufacturing and loading of assembly are performed by the same facility, the requirement of 3.6 is waived and the residual magnetism performed on the loaded assembly only.

6.3 Special instructions covering the modification, repair and assembly of remanufactured components shall be as follows:

6.3.1 Protective finish.- Remanufactured components which include head, body and trunnion shall be stripped of protective finish and re-plated in accordance with the applicable drawings and specifications.

6.3.2 Nylon pellets.- Bodies which contain nylon pellets and are to be re-plated, shall have the pellets removed and new pellets installed after re-plating in accordance with the applicable drawing.

6.3.3 Modification.

6.3.3.1 The modification of the body assembly shall be in accordance with figure 2.

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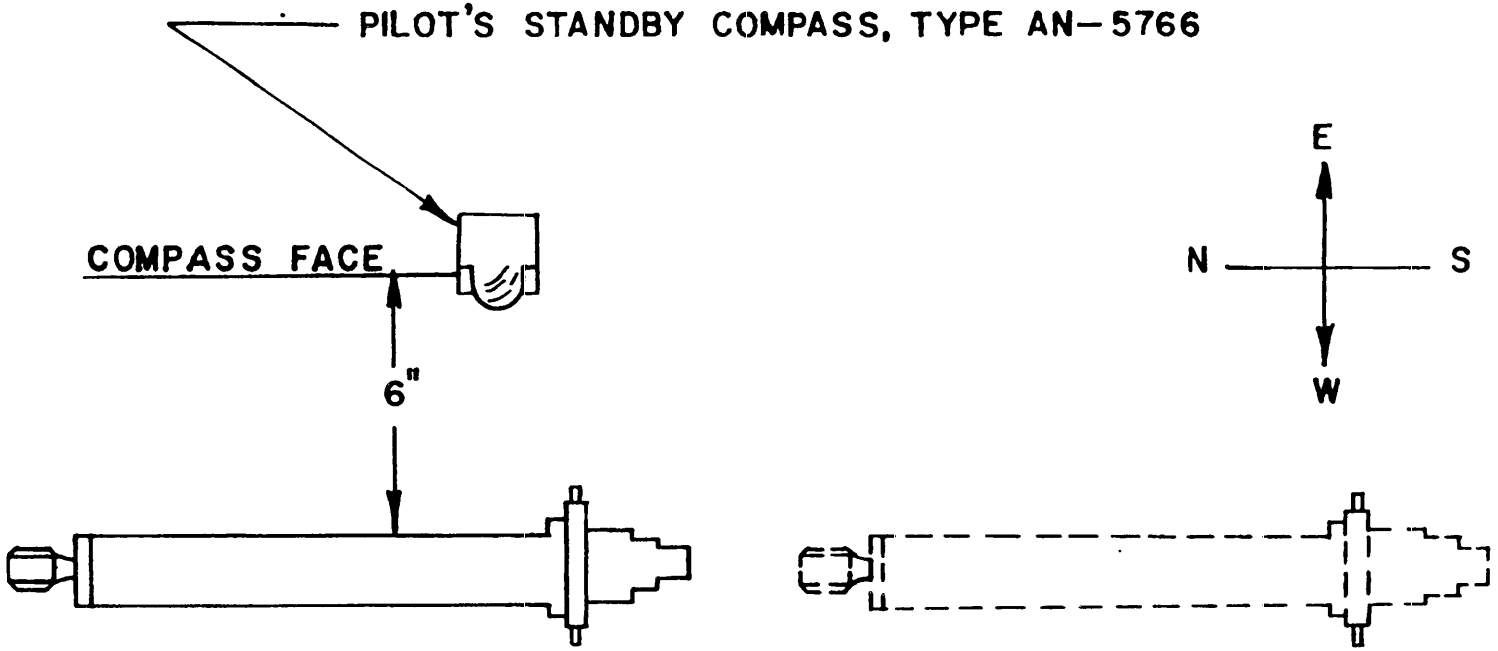
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PLAN VIEW

NOTE:
THRUSTER AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1. LAYOUT FOR DETERMINATION FOR RESIDUAL MAGNETISM IN THRUSTER, CARTRIDGE ACTUATED, M25

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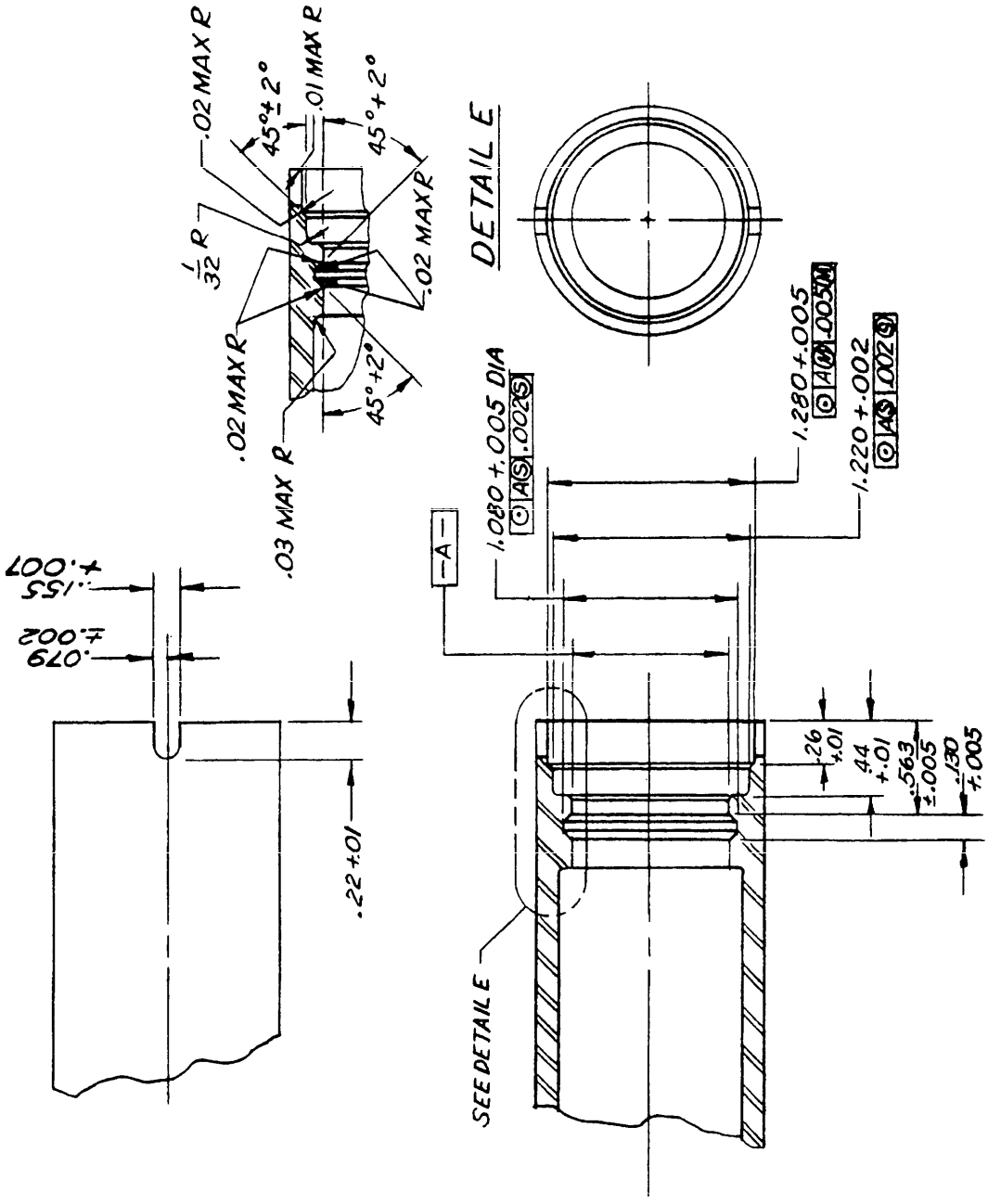


FIGURE 2

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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